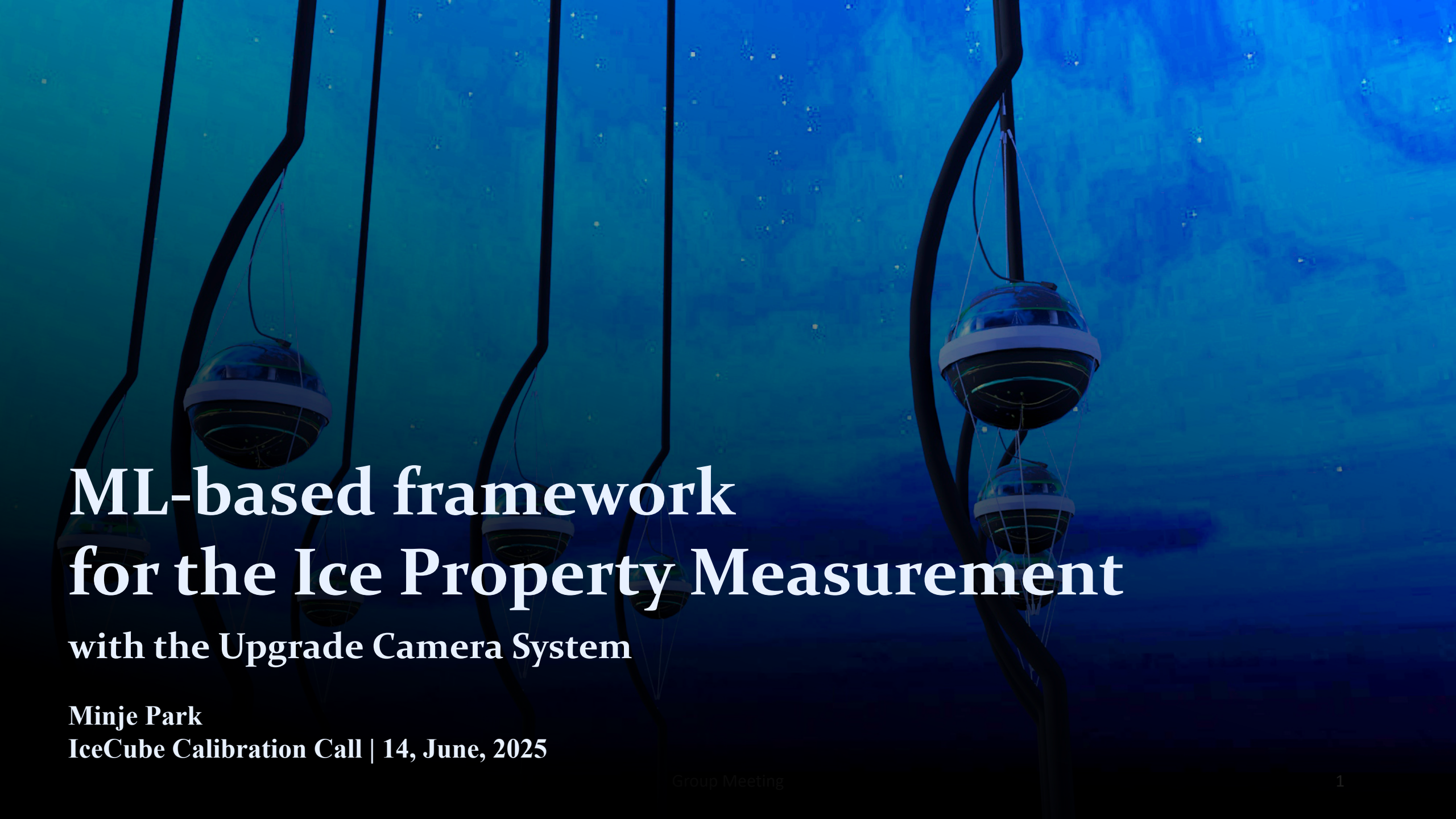




ML-based framework for the Ice Property Measurement

with the Upgrade Camera System

Minje Park

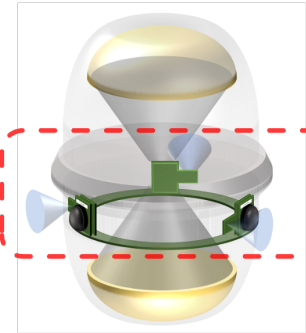
IceCube Calibration Call | 14, June, 2025

Upgrade Camera System and Camsim

Upgrade Camera System

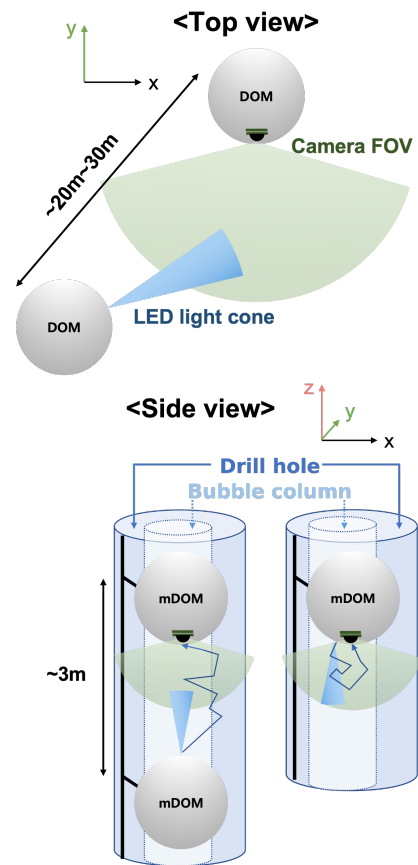
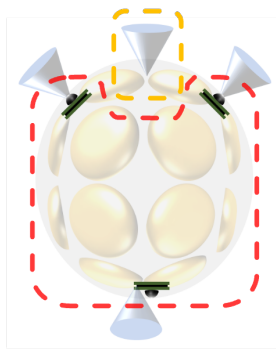
D-Egg

3×  + 
Camera LED

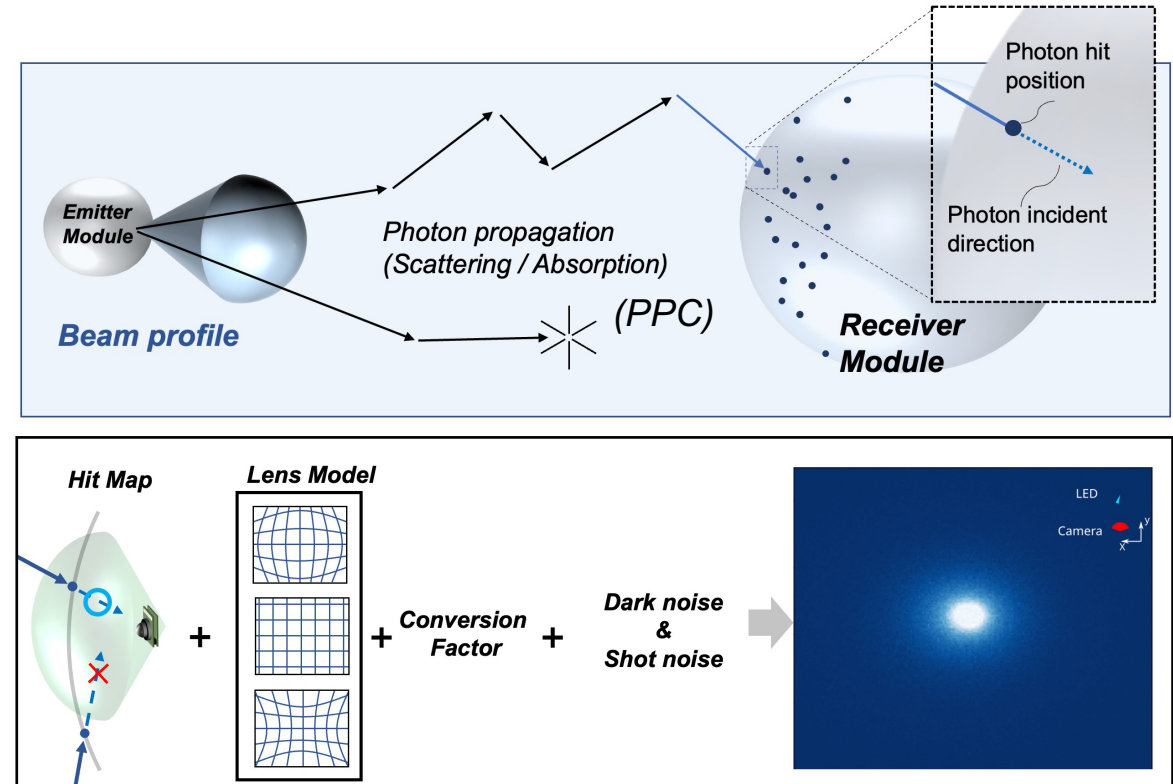


mDOM

3×  + 
1× 

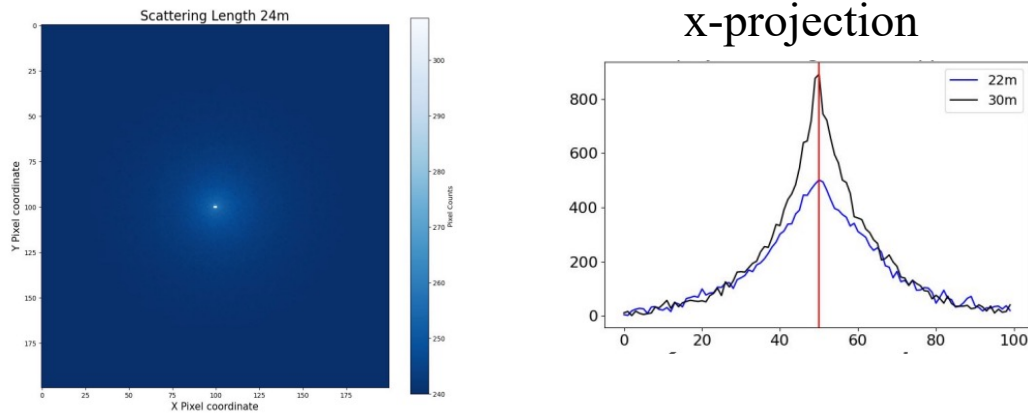


The CamSim



Previous studies

Jiwoong Lee, “Bulk Ice Simulation” @18AUG2023 Calibration Call ([link](#))

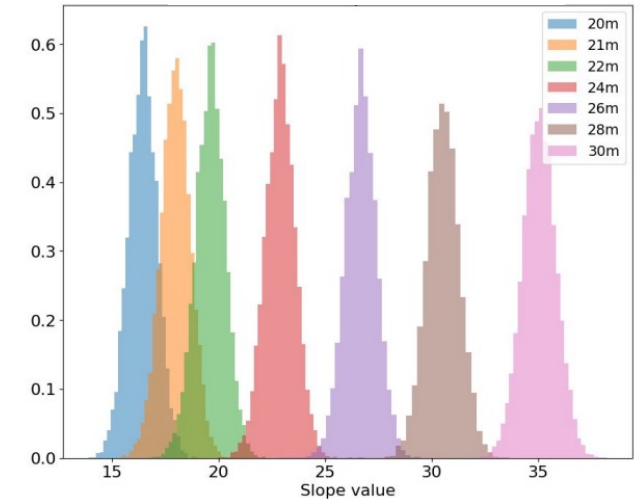


slope =

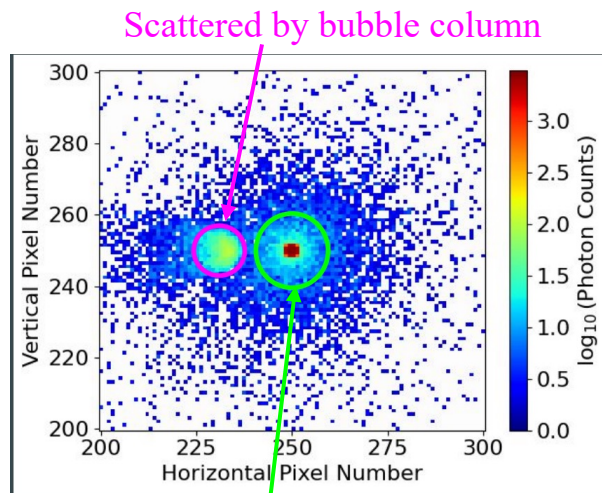
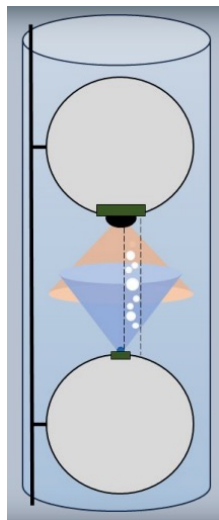
$$\left\langle \left| \frac{dy}{dx} \right| \right\rangle \equiv \frac{\sum_0^{max} \frac{dy}{dx} - \sum_{max}^{end} \frac{dy}{dx}}{\text{total bin number}}$$

Bulk ice λ_{scat} : 20~30m

Slope



Minji Shin, “Hole Ice Study” @18AUG2023 Calibration Call ([link](#))

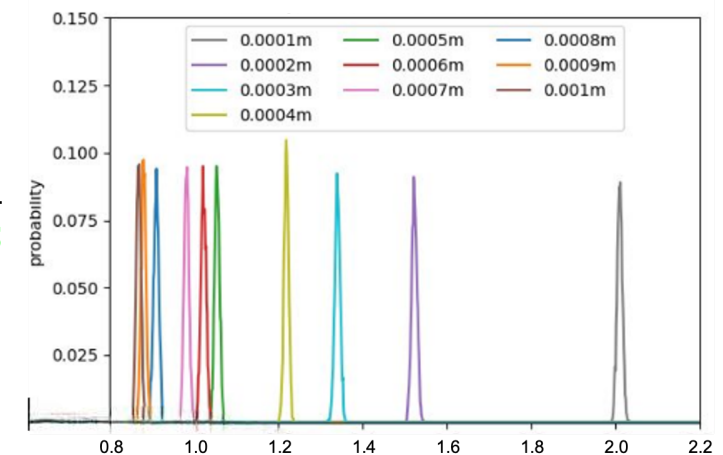


peak ratio =

$$\frac{\sum \text{scattered light photon counts}}{\sum \text{unscattered light photon counts}}$$

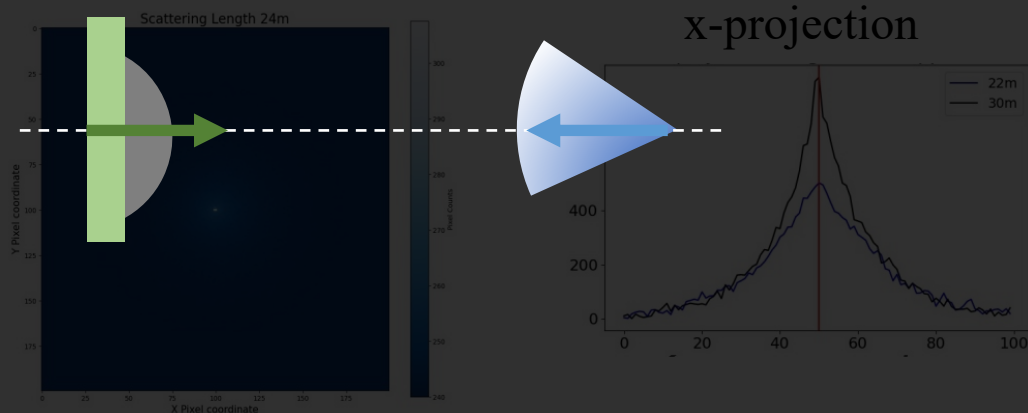
Bubble column λ_{scat} : 0.01~0.1cm

Peak ratio



Likelihood-based studies

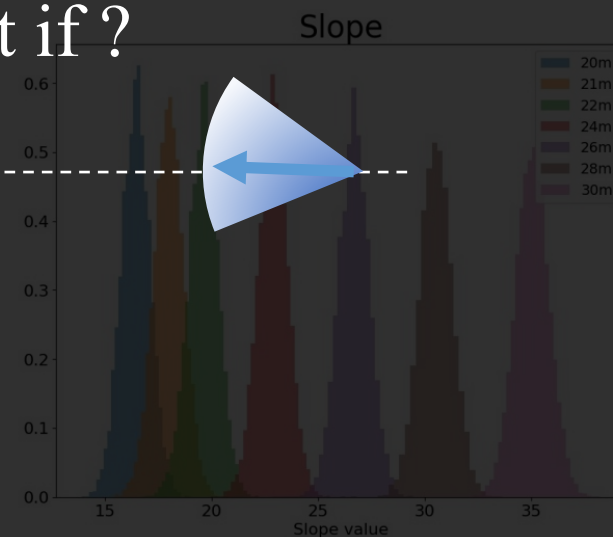
This work



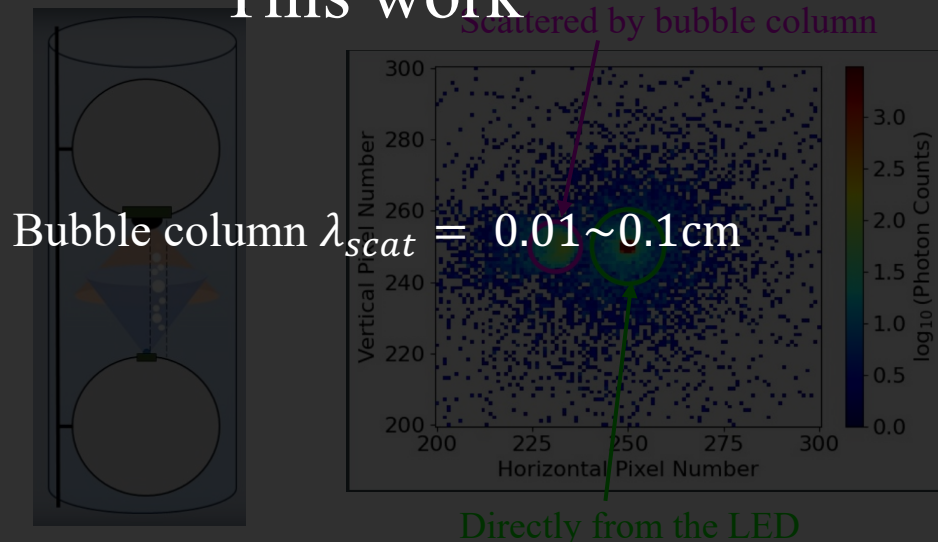
slope value =

$$\left\langle \left| \frac{dy}{dx} \right| \right\rangle \equiv \frac{\sum_0^{\max} \frac{dy}{dx} - \sum_{\max}^d \frac{dy}{dx}}{\text{total bin number}}$$

What if ?



This work



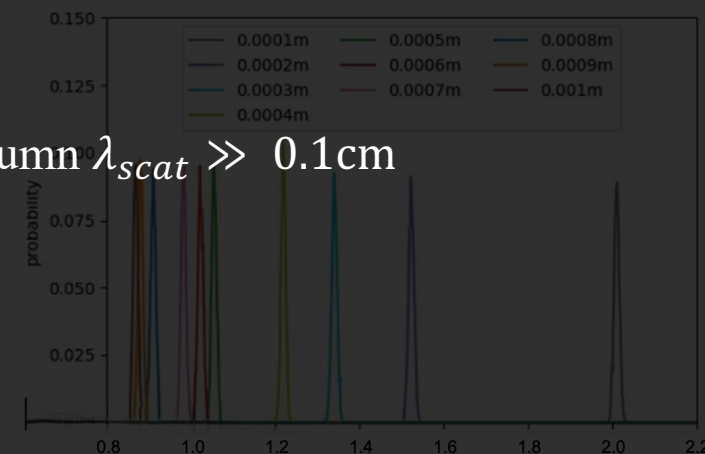
peak ratio =

$$\frac{\sum \text{scattered light photon counts}}{\sum \text{unscattered light photon counts}}$$

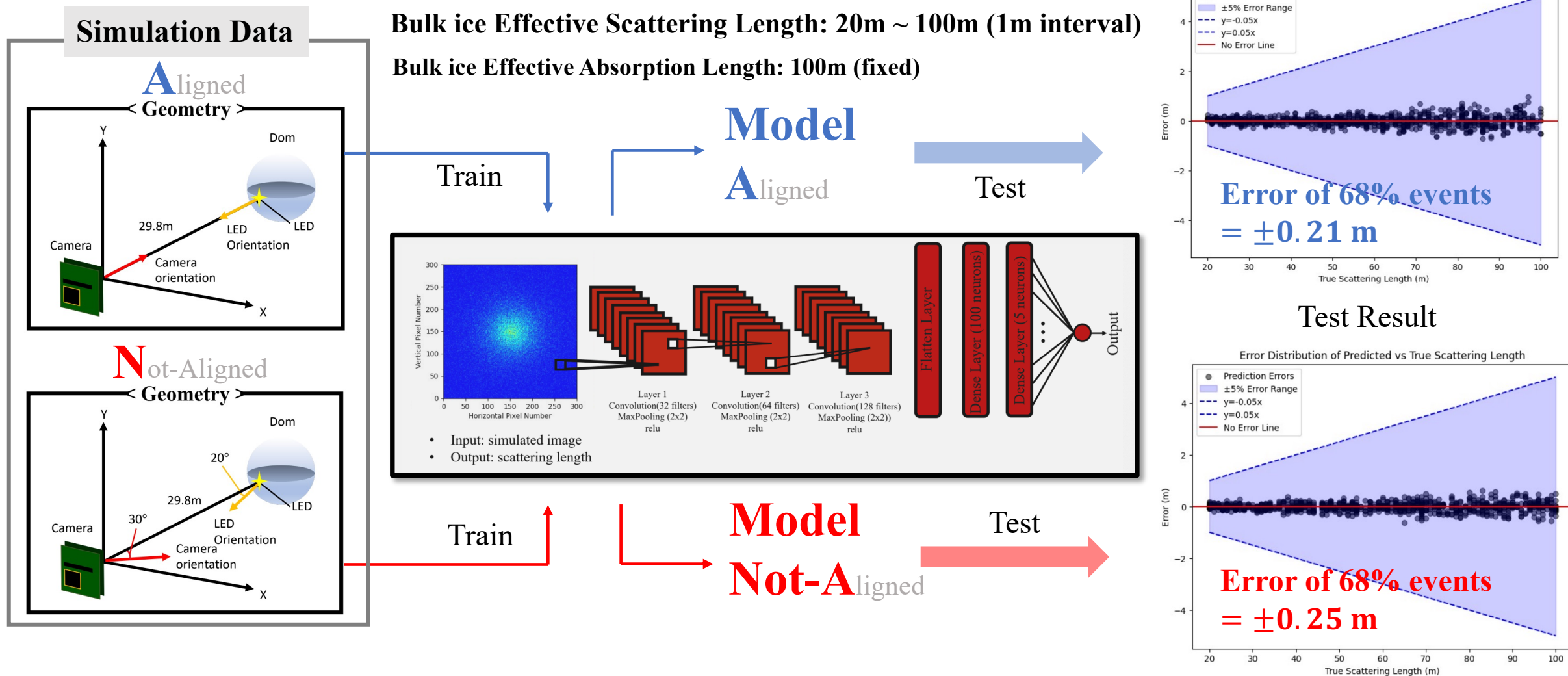
Bubble column $\lambda_{\text{scat}} \gg 0.1\text{cm}$

What if ?

Peak ratio



CNN-based bulk ice scattering length model

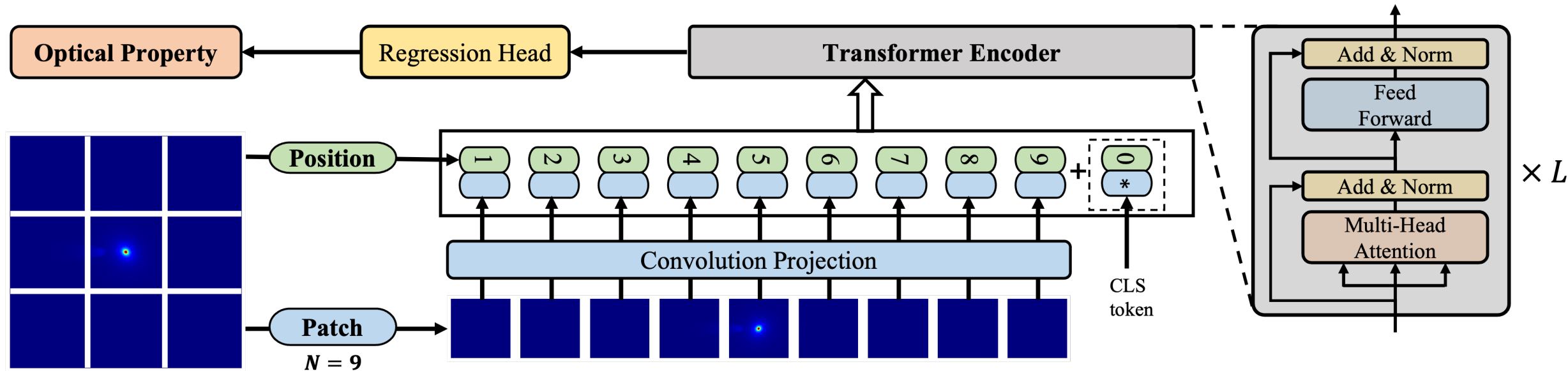


Still Separate Model for Separate configuration

What we want is model that can't work on various configuration,

OPTICUS: Architecture

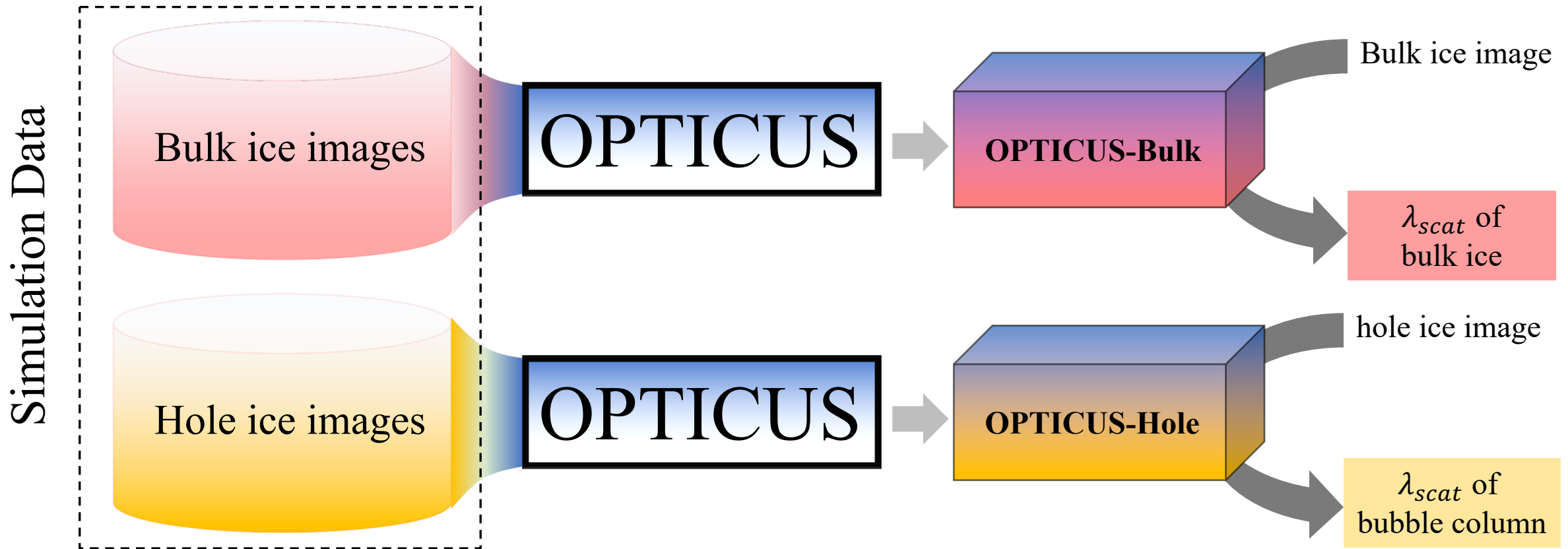
- CNN (Convolutional Neural Network) : extracting subtle and spatially distributed features
- Transformer-based architectures (ViT in vision task) : capturing global contextual relations across the image



- Developed **Optical Property Transformer for IceCube Upgrade Camera System, OPTICUS**

OPTICUS: Bulk & Ice

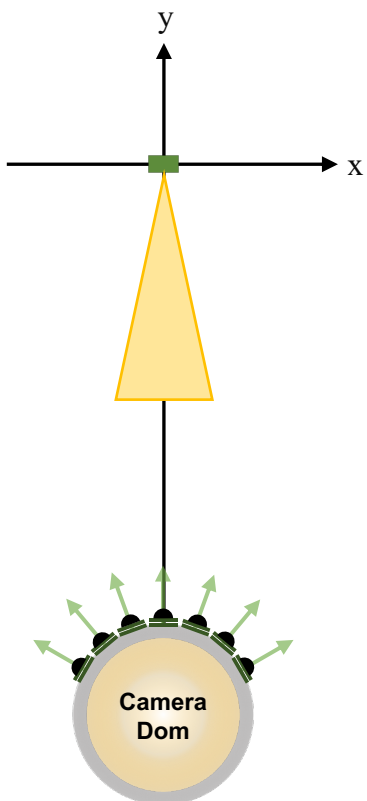
- To test the OPTICUS, we built two models, the OPTICUS-Bulk and the OPTICUS-Hole
 - OPTICUS-Bulk : predicting scattering length of the bulk ice
 - OPTICUS-Hole : predicting scattering length of the bubble column



Note : Although OPTICUS-Bulk & Hole has same architecture two models has different hyper parameter

Bulk ice Simulation Data

Geometric Setting



LED

- Position : (0, 29.8, -2200)
- Orientation : 270 °
- Nphoton : 10^{12}
- Wave length : 470nm

Camera

- Position : (0, 0, -2200)
- Orientation : 7 cases
 $> 30^\circ \sim 150^\circ$, step = 20°
- Aperture : 20
- Exposure time : 10s

Total 7 Geometric Configurations

Ice Properties Setting

For each geometric configuration vary

- Bulk Ice Scattering Length : 20m ~100m (1m interval)
 $\Rightarrow$ 81 cases

Total 567 Simulation Images

Camera Noise

For each simulation image

- Based on Noise Model, generate 100 noisy image

Total 56,700 Images

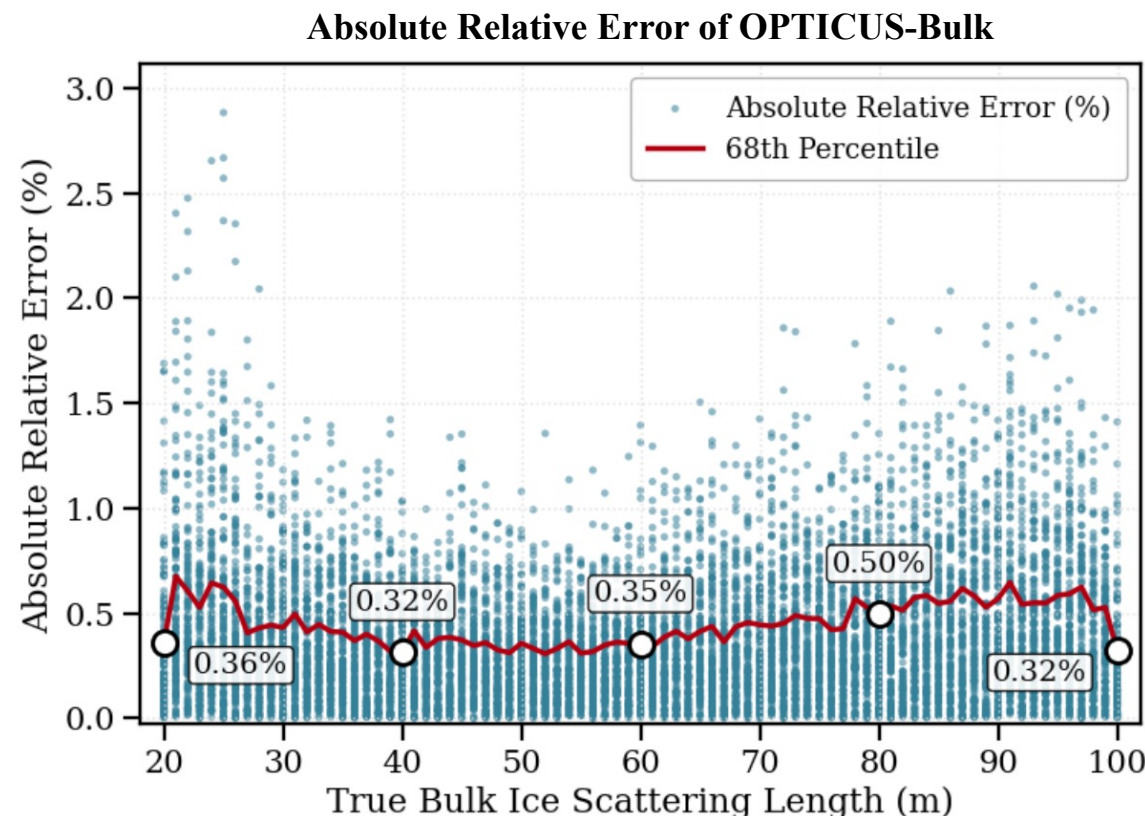
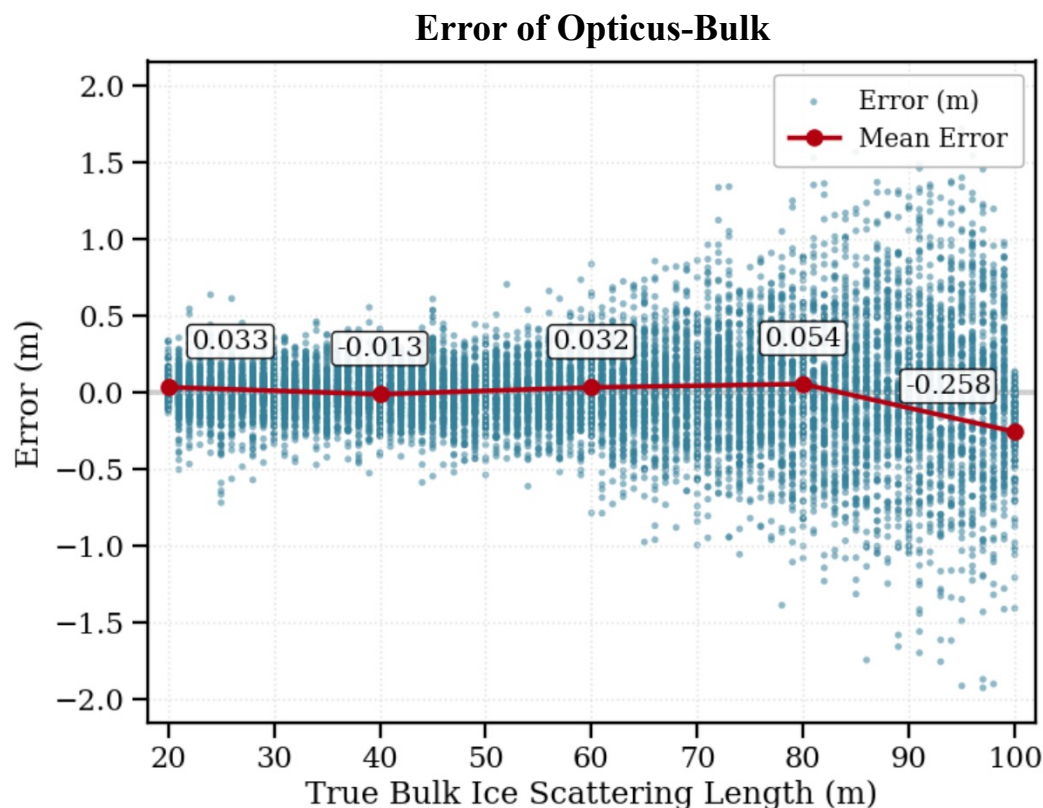
Train Image	Test Image
51,030	5,670

Test-Performance

Plot included
in Proceeding

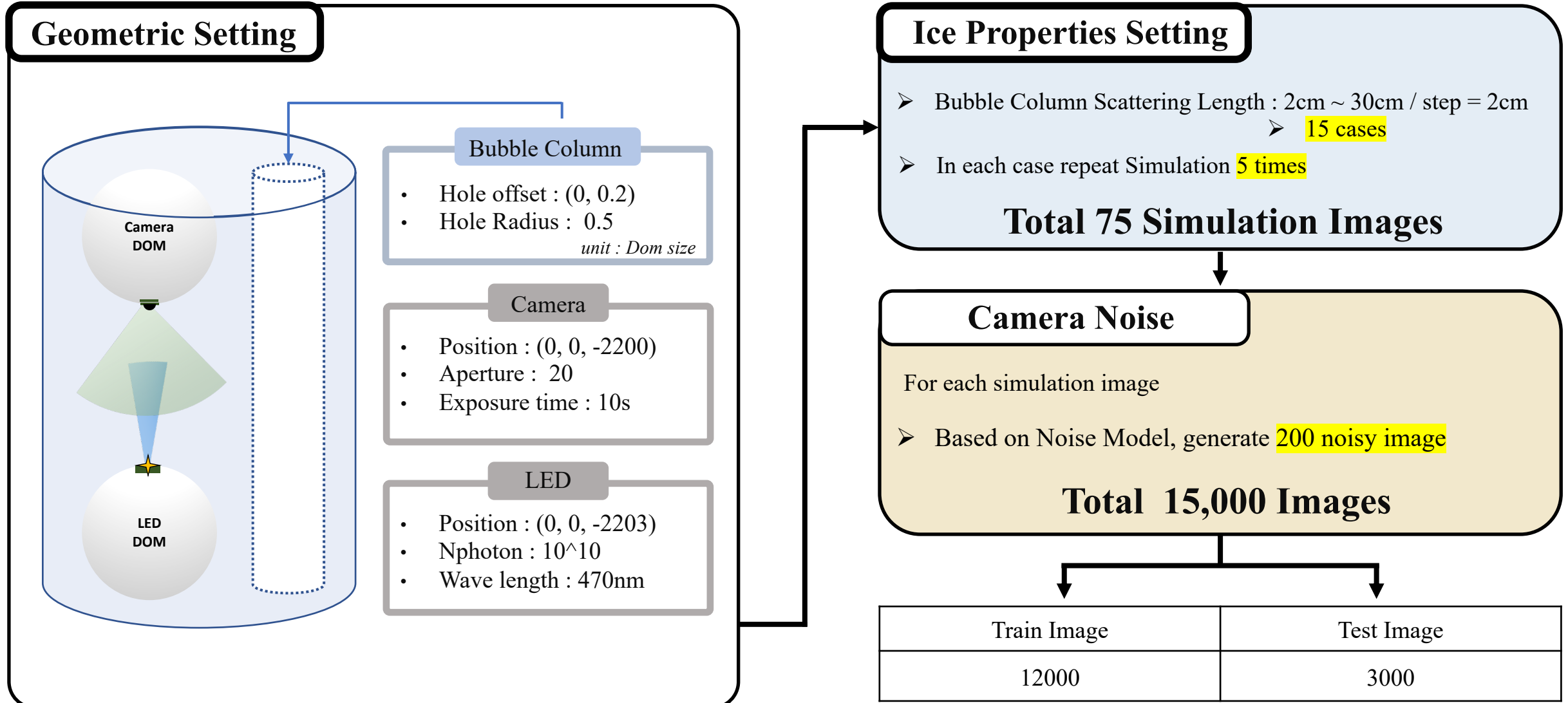
- Test the performance of **OPTICUS-Bulk**, 10% of the simulated images were used as test data.
- The evaluation was conducted by feeding the trained model with the test data (simulation images) as input and **comparing the predicted bulk ice scattering lengths with the ground truth**.
- The error was defined as: **error = true (simulation) – predicted value**

$$\text{Relative Error}(\%) = \frac{|\text{Error}|}{\text{true scattering length}} * 100$$



***Note:** During testing, the camera orientation was **not** explicitly provided to the model. This was intentional, to verify whether **OPTICUS-Bulk** can infer the scattering length **regardless of the camera orientation**, based solely on the image input.

Hole ice Simulation Data



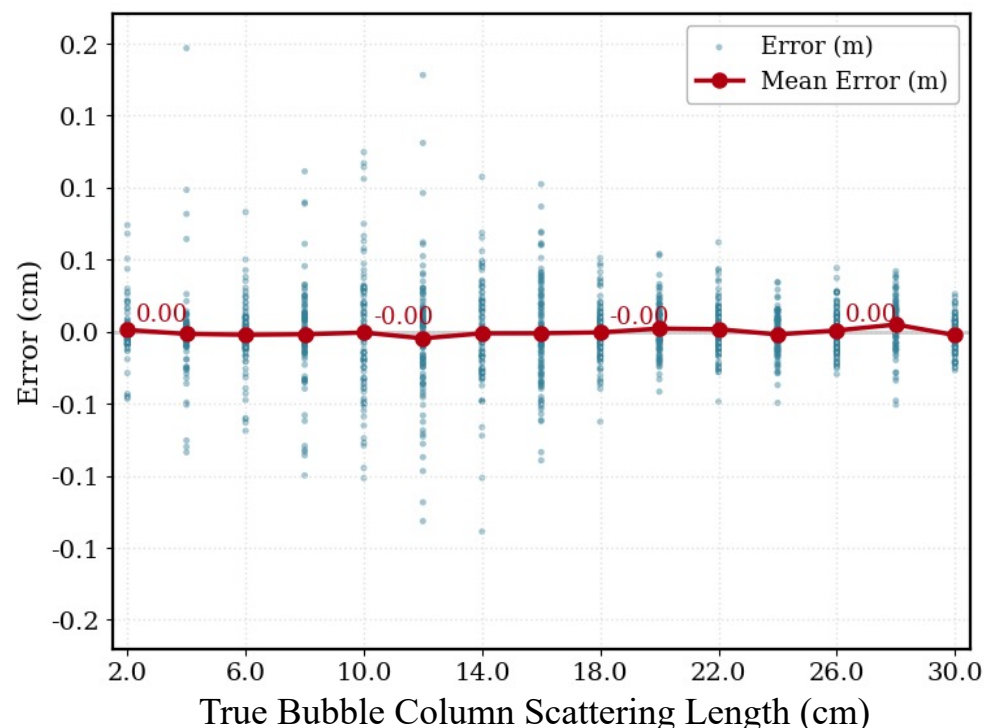
OPTICUS-Hole / Test Performance

Plot included
in Proceeding

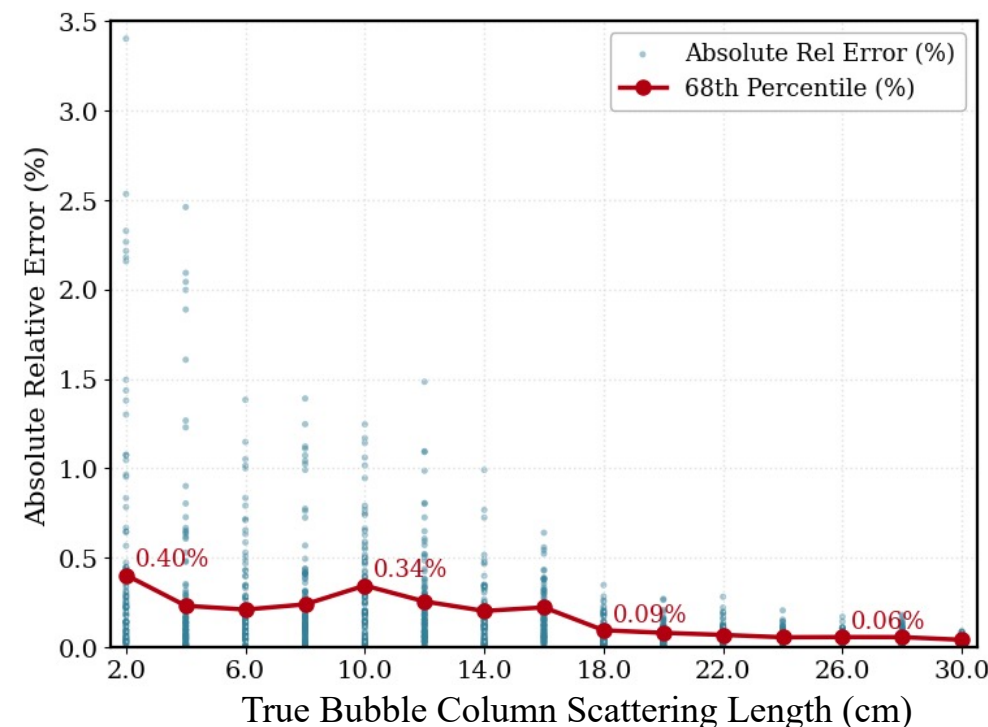
- To test the performance of **OPTICUS-Hole**, 10% of the simulated images were used as test data.
- The evaluation was conducted by feeding the trained model with the test data (simulation images) as input and **comparing the predicted bubble column scattering lengths with the ground truth**.
- The error was defined as: **error = true (simulation) – predicted value**

$$\text{Relative Error}(\%) = \frac{|\text{Error}|}{\text{true scattering length}} * 100$$

Error of Opticus-Hole



Absolute Relative Error of OPTICUS-Hole



Points of Caution

1. Limited variation: Only scattering length was changed

- This study serves as an **initial test** to confirm that our new architecture performs well in predicting ice properties from images.
- However, only **one optical parameter** (scattering length) was varied per model.
 - This makes the learning task simpler, resulting in **better-than-expected sensitivity**.
- In future studies, we will vary **multiple parameters** simultaneously.
 - Accounting for **inter-parameter correlations** may lead to reduced model sensitivity.

2. Simplified simulation setup

- The model was trained solely on simulation data, which may not fully represent real-world conditions.
- **Improving the realism of simulations** is essential for more accurate sensitivity studies.
- For this, we are currently working on **CamSim updates**,
 - e.g., the Utah team is preparing **water-tank measurements** to refine the camera model.

Conclusion

- The primary goal of the Upgrade Camera System is to **measure ice properties** from captured images.
- In addition to likelihood-based methods, we developed a **Machine Learning–based** framework.
- Following our **previous CNN-based** approach, we designed a **transformer-based architecture** called **OPTICUS**.
- To evaluate the performance of OPTICUS and the camera’s measurement sensitivity, we **trained and tested** the model using **simulated images generated by CamSim**.
- Depending on the measurement target, we constructed two models as below.

	OPTICUS-Bulk	OPTICUS-Hole
Purpose	Predicting bulk ice scattering length	Predicting bubble column scattering length
Simulation Images for training/testing	Simulated images were created by fixing the LED source and varying the camera orientation across different bulk ice scattering lengths .	Both LED and camera were fixed . Simulations varied the scattering length of the bubble column , while keeping its size and position constant .

- Both models demonstrated **robust performance across the test data**, highlighting the potential of OPTICUS for characterizing ice properties. (OPTICUS-Bulk error: ~ 0.4%, OPTICUS-Hole error: ~ 0.5%)
- As a foundation for the final model, future work will include **additional ice parameters** and development of CamSim to generate more **realistic training data**.

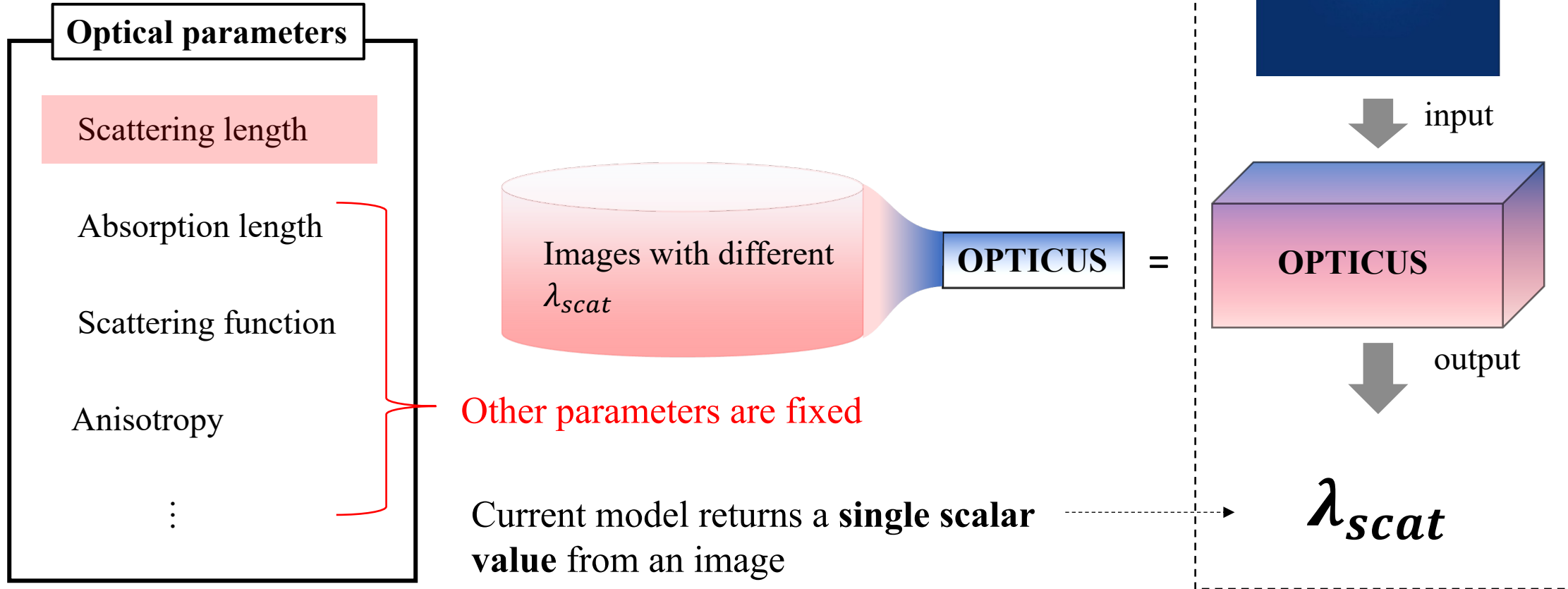


Thank You for your Attention

You can contact me with: pmj032400@naver.com

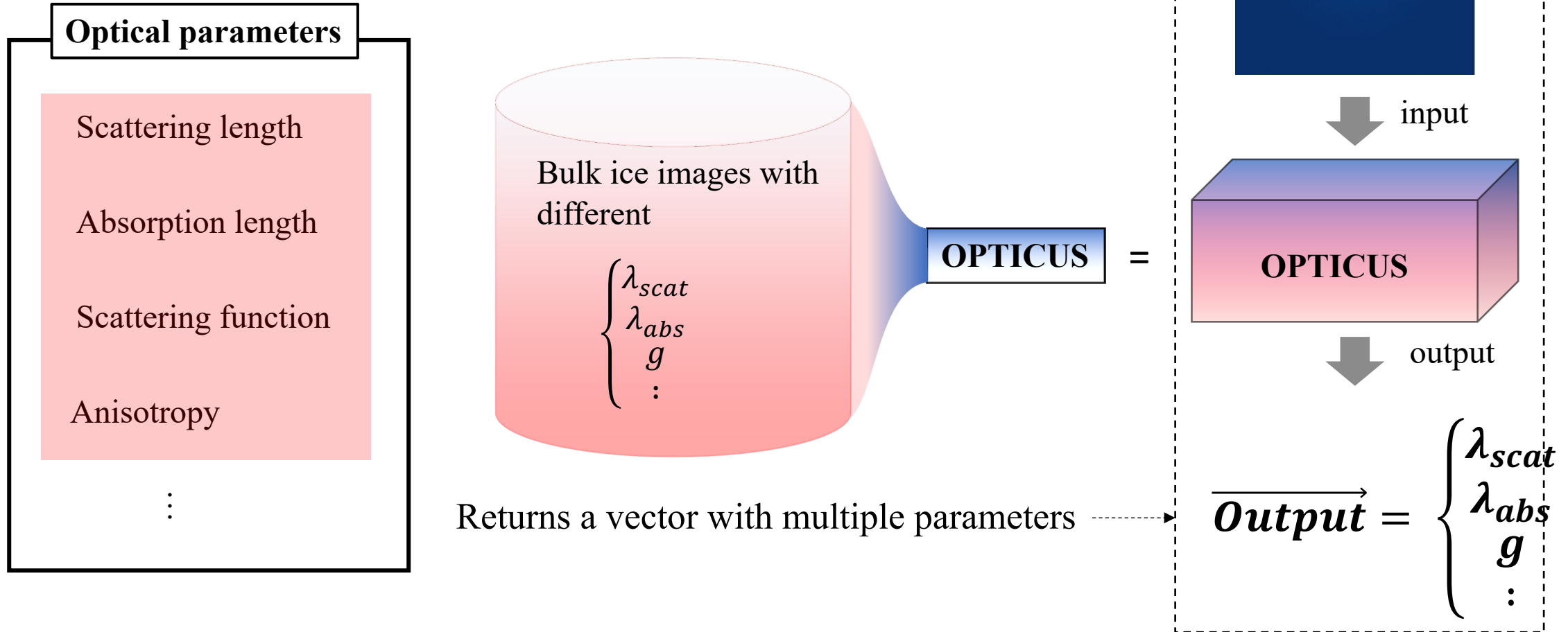
OPTICUS : Current status

- As a blueprint to construct ultimate model, we first test with varying the scattering length only.

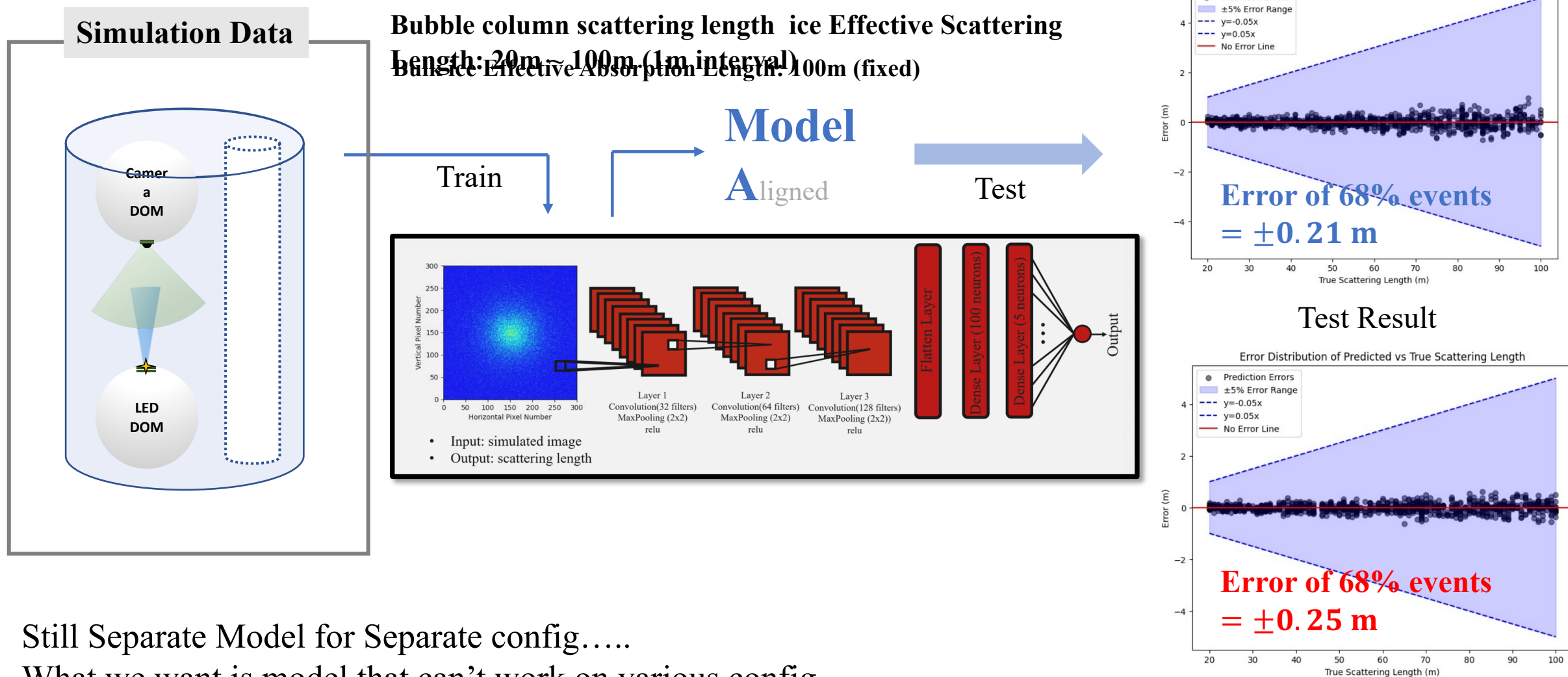


OPTICUS : Goal

- The ultimate goal is to build a model that can infer multiple parameters based on a image



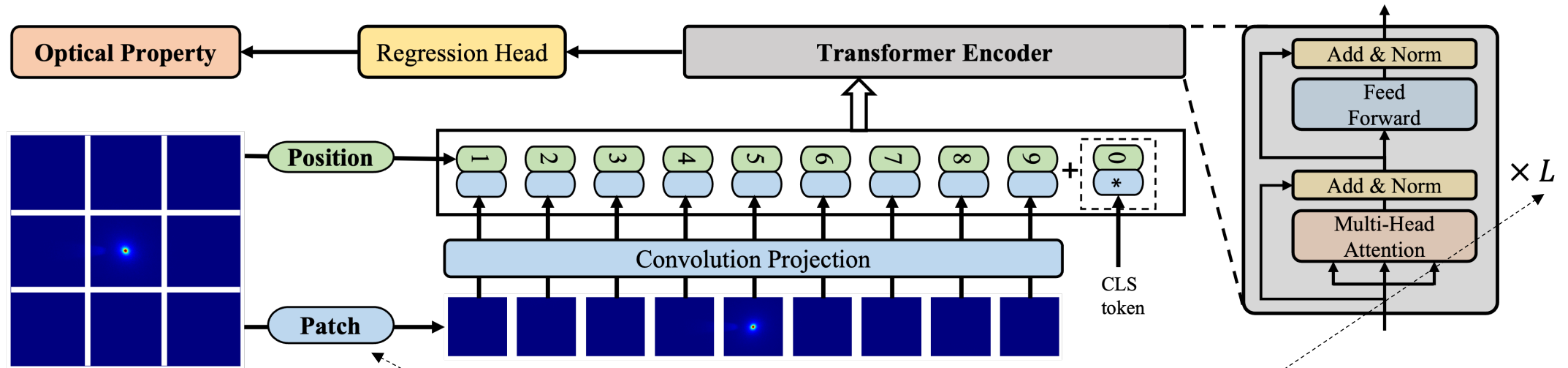
CNN-based hole ice scattering length model



Still Separate Model for Separate config.....

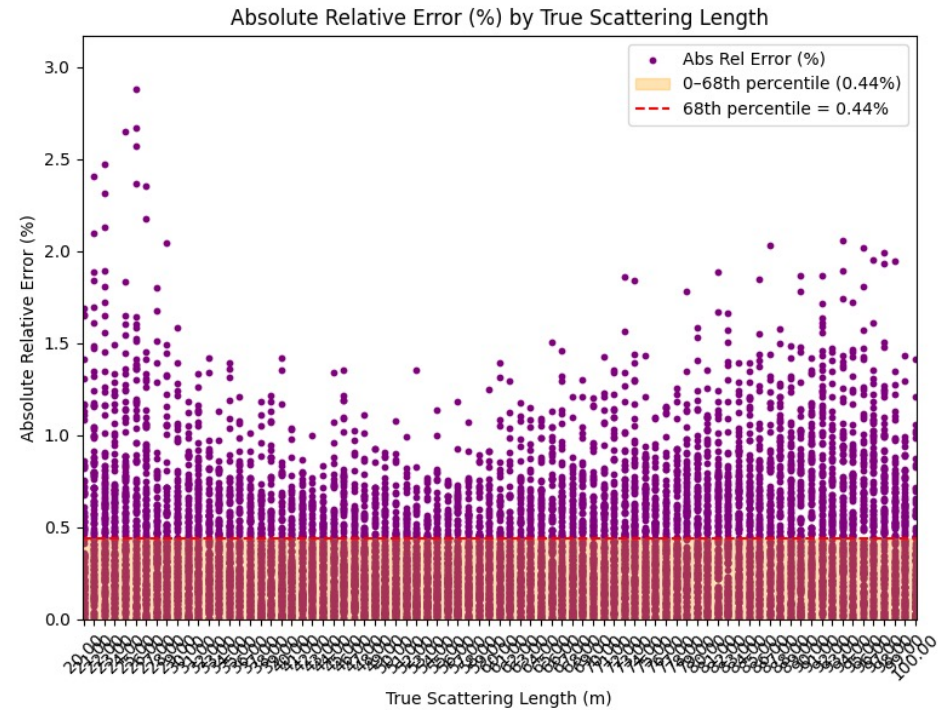
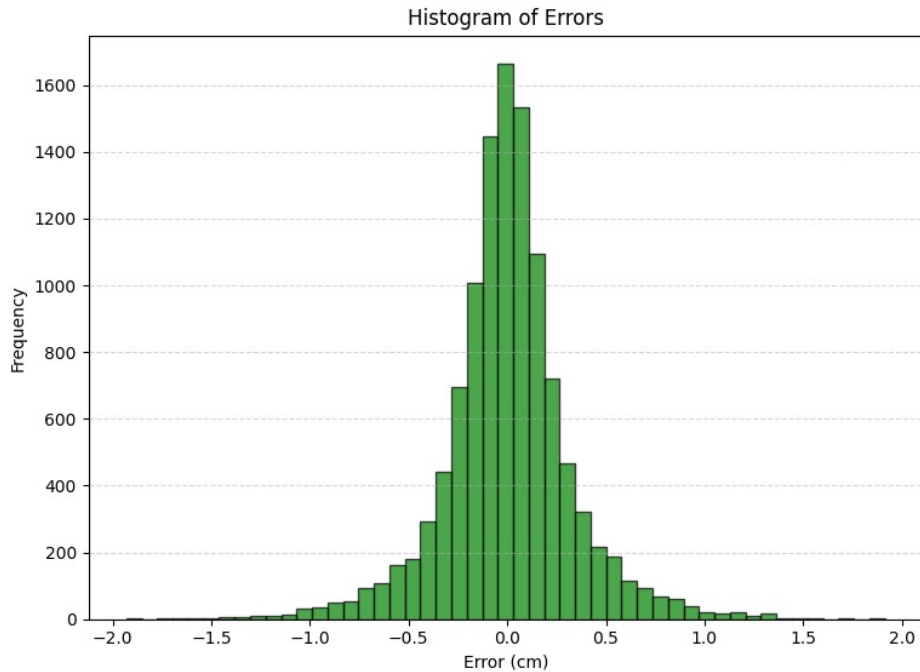
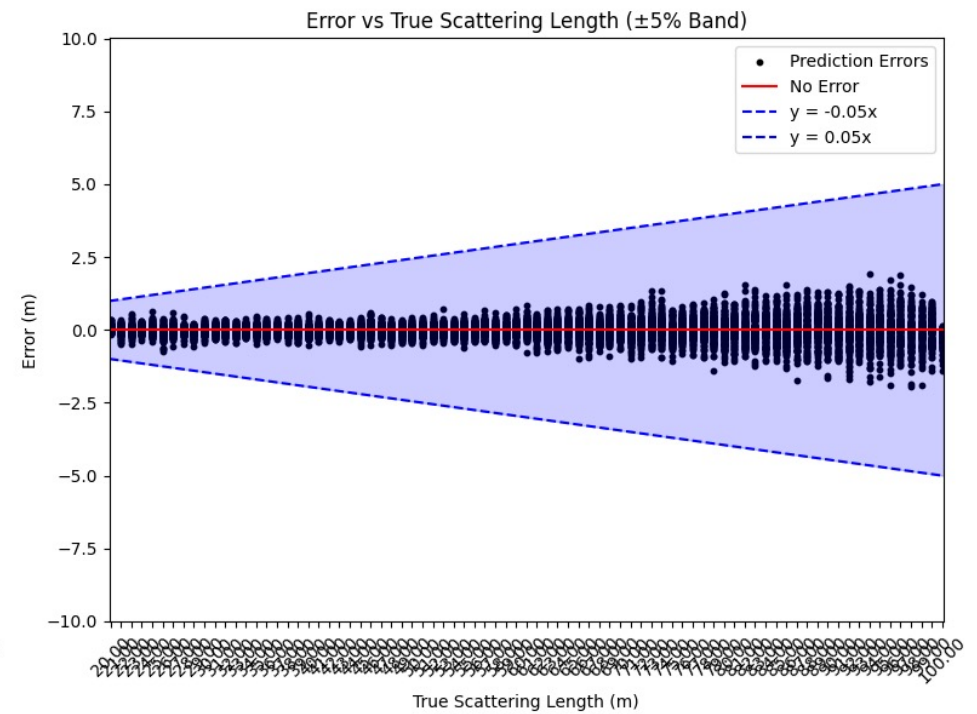
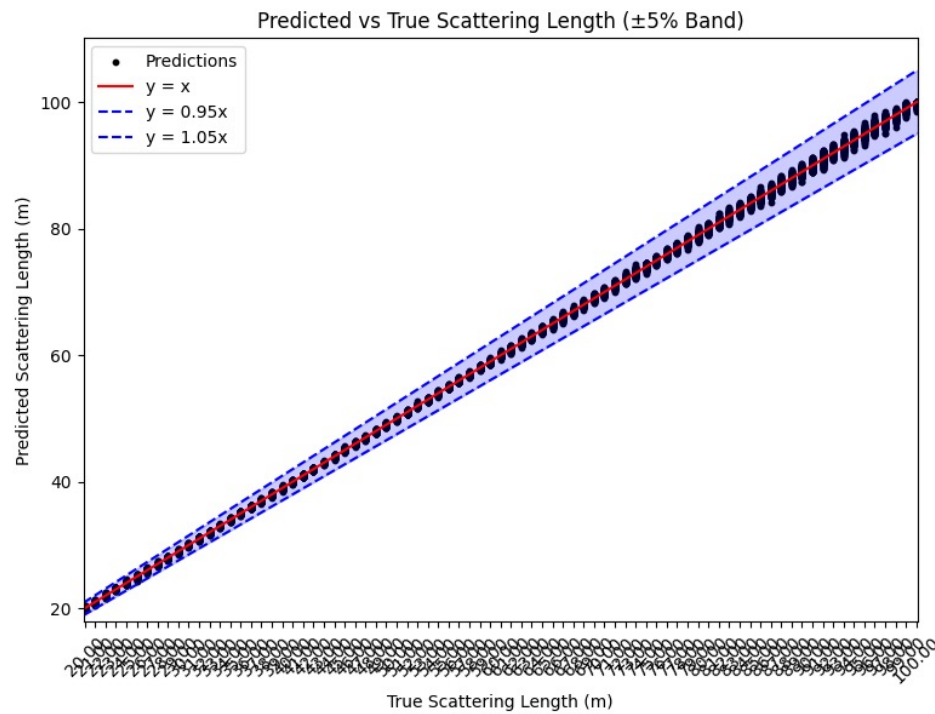
What we want is model that can't work on various config,

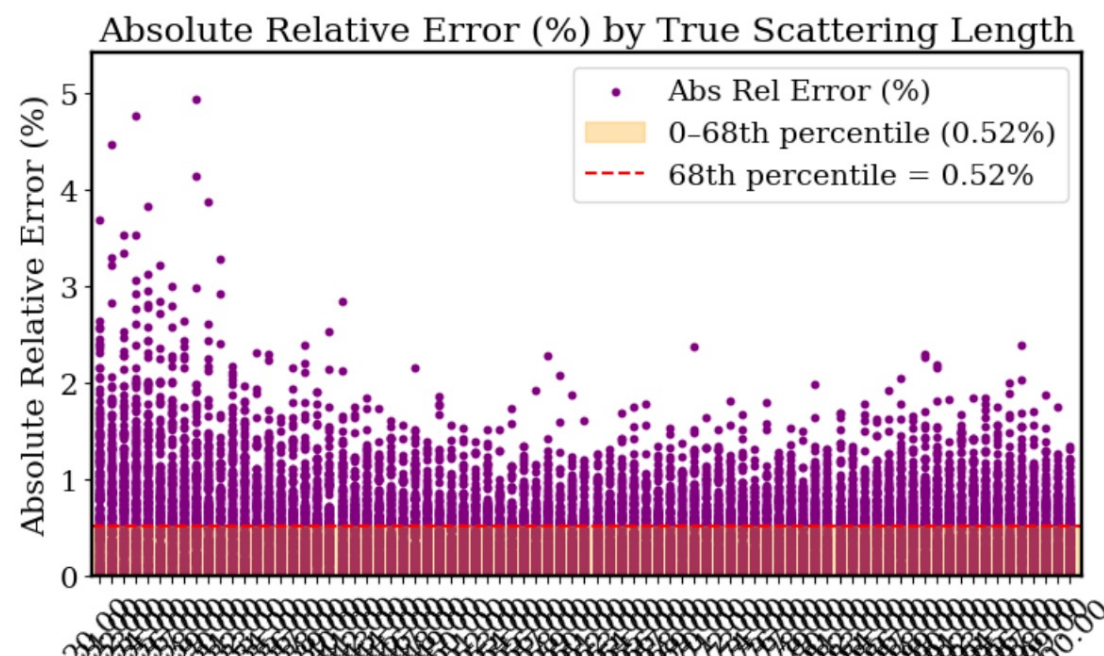
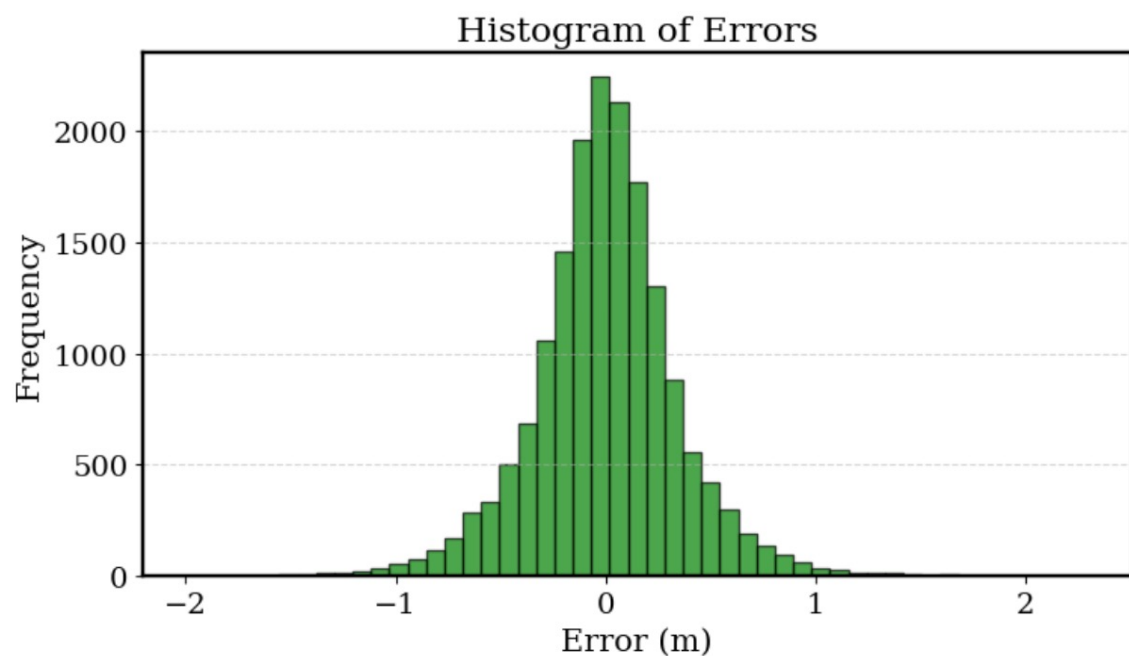
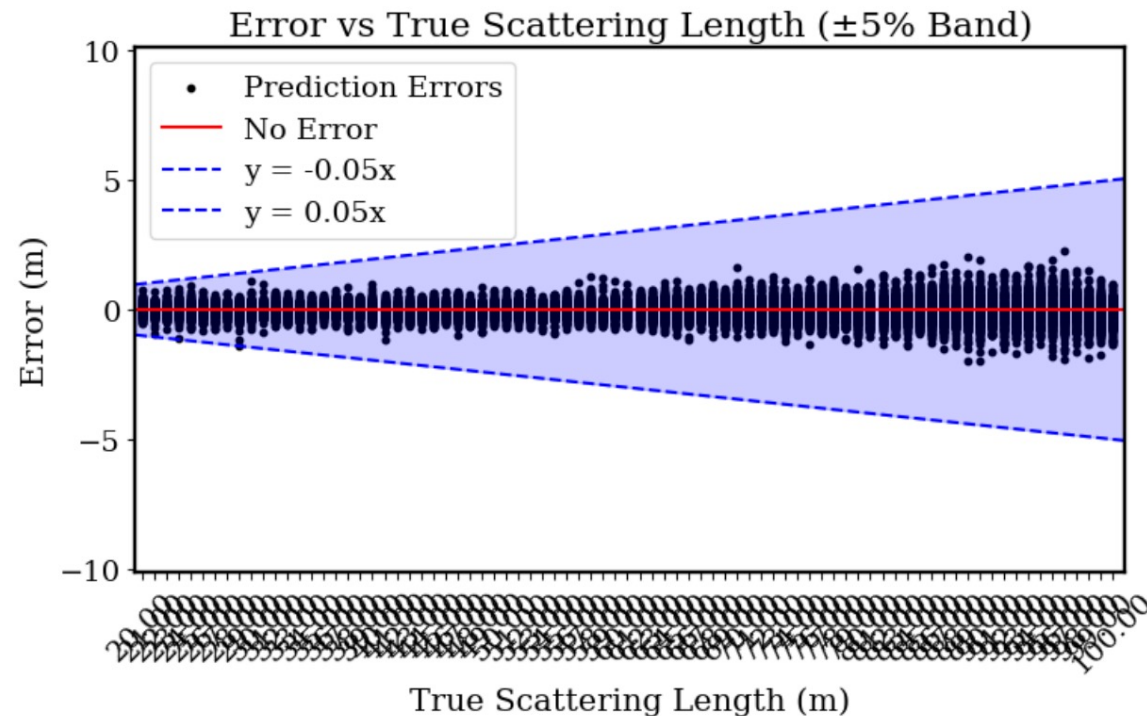
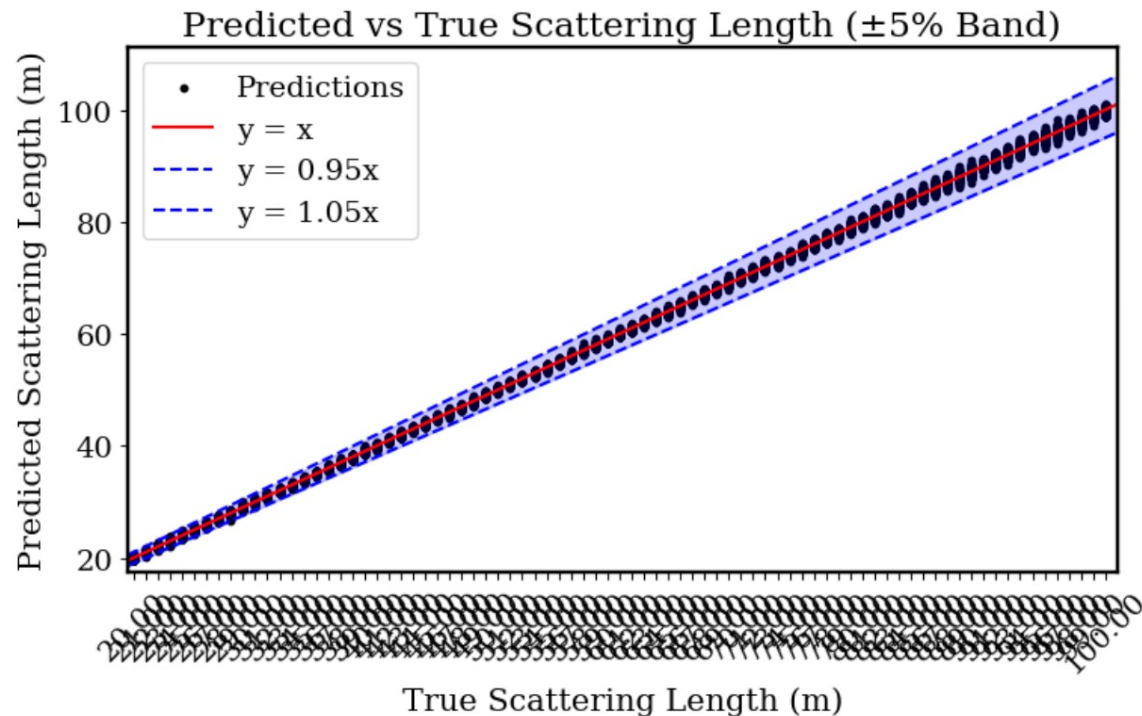
OPTICUS: Difference in model parameters



OPTICUS-Bulk	Patch size = 30×30	$L=5$
OPTICUS-Hole	Patch size = 50×50	$L=3$

1. OPTICUS OPTI





1. OPTICUS OPTI

